Serotype-Driven Dynamics of Dengue R_t in Singapore

A Semiparametric Bayesian Renewal Equation Model with Gaussian Process Residuals

Ansel Lim

2026-04-16

Why Decompose Dengue R_t?

The problem:

- Singapore: recurrent dengue epidemics, year-round transmission
- Multiple interacting drivers: climate, vector ecology, serotype cycling, population immunity
- Standard R_t methods (EpiEstim, EpiNow2) estimate Rt as a single latent quantity, without attributing variation to specific drivers

Our approach:
Decompose $\log(R_t)$ into **measured** (climate) and **unmeasured** (residual GP) components $\rightarrow$ examine residual for serotype dynamics

Why Gaussian Processes?

- **Nonparametric flexibility**: kernel encodes prior beliefs about smoothness and length scale without fixing a functional form
- **Additive decomposition**: climate covariates + GP residual in one coherent Bayesian model (sum of GPs is a GP)
- **Interpretable hyperparameters**: ρ (weeks) and α (log-R_t scale) carry direct epidemiological meaning
- **Calibrated uncertainty**: credible intervals tighten near data and propagate through the renewal equation

Research Questions

1. **Decomposition & accounting.** How is weekly $\log R_t$ variation partitioned between *climate covariates* and the *residual Gaussian process component*, and how many cases does the residual component account for over the study period relative to a climate-only accounting? *(variance decomposition and retrospective residual decomposition)*

2. **Serotype dynamics.** Is variation in the residual Gaussian process temporally coincident with serotype replacement events, and what lag and periodicity structures characterize the residual series? *(switch profiles, cross-correlation, spectral analysis, per-switch residual contribution)*

3. **Robustness & validation.** Are conclusions robust to climate lag choice, GP hyperprior means, and Gaussian process kernel specification? *(climate lag sweep, GP hyperprior sweep, kernel sensitivity)*

4. **Operational utility.** Can the Gaussian process framework provide skillful short-horizon forecasts of R_t? *(forecast scoring against a random walk baseline)*

5. **Outbreak risk.** How do R_t fluctuations translate to outbreak sustaining probability, and do serotype switches open windows of elevated vulnerability? *(branching process theory)*

Data Sources

Dengue cases

- Weekly notified cases, Singapore
- 2012–2022
- MOH Weekly Infectious Diseases Bulletin (data.gov.sg)

Climate covariates

- Weekly mean temperature (°C)
- Weekly total rainfall (mm)
- Standardized, **4-week lag** (vector ecology + case ascertainment; climate share $\leq 3\%$ across lags 2–6 wks)
- Meteostat API (Changi Airport)

Serotype surveillance

- Monthly DENV-1–4 proportions (NEA)
- GAM multinomial logistic smoothing
- Shannon entropy & switch events
- Used **post-hoc only** (not as model covariate)

Post-hoc design avoids:

- Circularity / endogeneity
- Post-treatment conditioning
- Sampling bias (non-random subset typed)
- Measurement error (small-n monthly)
- Resolution mismatch (monthly vs weekly)
- Underpowered regression ($n = 7$ switches)
- Forced alignment with residual GP

Model: Semiparametric Decomposition of $\log(R_t)$

$$\log(R_t) = \underbrace{\mu}_{\text{intercept}} + \underbrace{\beta_{\text{temp}} \cdot \text{temp}_{t-4} + \beta_{\text{rain}} \cdot \text{rain}_{t-4}}_{f_{\text{climate}}} + \underbrace{f_{\text{residual}}(t)}_{\text{GP}}$$

Residual GP:

- Matérn 3/2 kernel (default in EpiNow2)
- Once-differentiable: allows rapid R_t changes
- **HSGP approximation** ($M = 110$ basis functions): $\mathcal{O}(N^3) \to \mathcal{O}(MN)$

Transmission model:

- Renewal equation: $\lambda_t = R_t \sum_{s=1}^{6} c_{t-s} w_s$
- Generation interval via **Monte Carlo convolution** of IIP + viremia + EIP + bite delay (Chan & Johansson 2012)

Observation model: Negative binomial

$S = 6$ weeks: captures essentially all the GI mass at the empirical mean weekly temperature (27.9 °C, GI mean 20.4 d) and the 97.5th percentile at 25 °C cool-period sensitivity (mean ≈ 25 d, EIP-driven long tail).

Inference: CmdStan 2.38.0, 4 chains × (1000 warmup + 1000 sampling). All $\hat{R} < 1.01$, zero divergences.

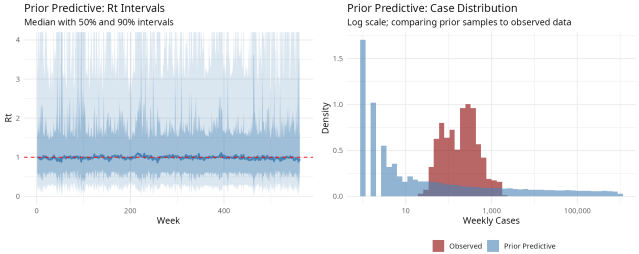
Weakly Informative Priors Generate Plausible Rt and Cover Observed Case Magnitudes

Parameter	Prior	Implication
μ	$\mathcal{N}(0, 0.5)$	Median $R_t = 1$
β_{temp}, β_{rain}	$\mathcal{N}(0, 0.5)$	Centered at no effect
$\log(\rho)$	$\mathcal{N}(\log 6, 0.5)$	Length scale $\sim$6 wks
$\log(\alpha)$	$\mathcal{N}(-1.2, 0.5)$	Amplitude $\sim$0.3
ϕ	half-$\mathcal{N}(0, 5)$	Overdispersion

How ρ, α were chosen:

- ρ: median $\approx 2\times$ generation interval ($\sim$3 wks). Fast enough to track outbreak transitions, long enough to avoid overfitting weekly noise. 95% CI $\sim[2.2, 16.1]$ wks spans outbreak-peak to seasonal scales.
- α: median ≈ 0.3 on $\log R_t$ scale $\Rightarrow$ residual alone multiplies R_t by $\exp(\pm 0.3) \approx [0.74, 1.35]$, matching realistic outbreak swings. 95% CI for $\alpha \approx [0.11, 0.80]$ admits weak and strong residual effects.

Prior predictive check: 500 draws $\rightarrow$ renewal equation $\rightarrow$ median $R_t \approx 1$; prior case distribution covers observed range.



Climate Explains Less than 1 Percent of R_t Variance

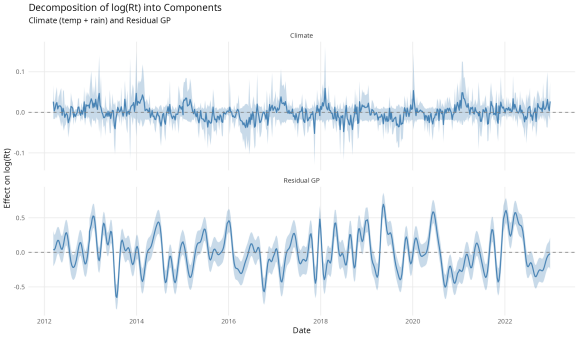
Climate effects are negligible:

- Temperature: $\times 0.983$ [0.962, 1.011]
- Rainfall: $\times 1.000$ [0.982, 1.020]
- Neither distinguishable from 1.0

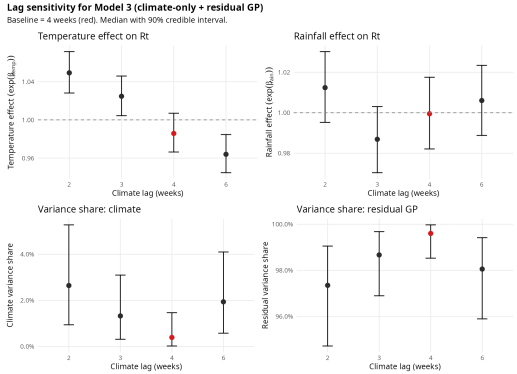
Variance decomposition:

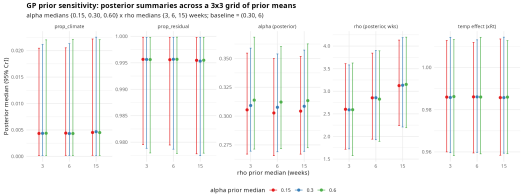
- Climate: **0.6%** of R_t variance
- Residual GP: **99.4%**

Interpretation: Climate is not negligible in general: it acts through slower vector-ecology mechanisms. At **weekly resolution**, faster processes dominate.



Robustness: Climate Lag and GP Hyperprior Sweep





Method — climate lag:

- Refit at climate lags 2 / 3 / 4 / 6 wks

Result:

- Climate var. share $< 3\%$ at every lag (2.6, 1.3, 0.4, 1.9%)
- Temp coefficient **flips sign** ($1.05 \rightarrow 0.96$): no stable weekly climate signal

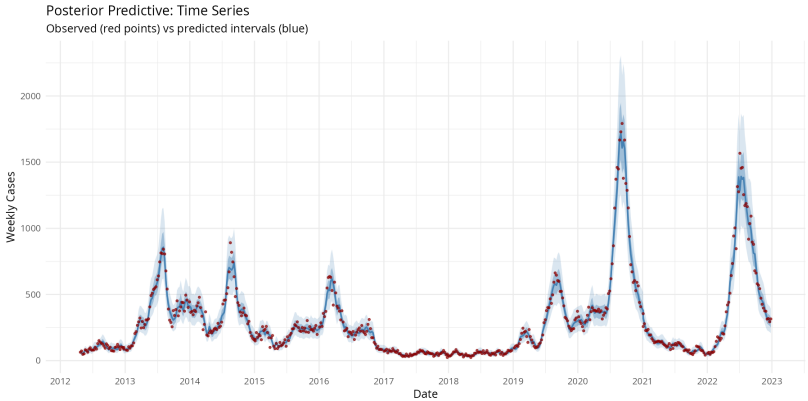
Method — GP hyperprior sweep: 3×3 grid of hyperprior combinations: α prior median $\in \{0.15, 0.30, 0.60\}$, ρ prior median $\in \{3, 6, 15\}$ weeks

Result:

- Posterior variance attributable to climate: **0.43–0.47%** across all 9 cells
- Posterior $\hat{\alpha} \approx 0.30$ regardless of prior median
- Posterior $\hat{\rho} \in [2.6, 3.2]$ weeks

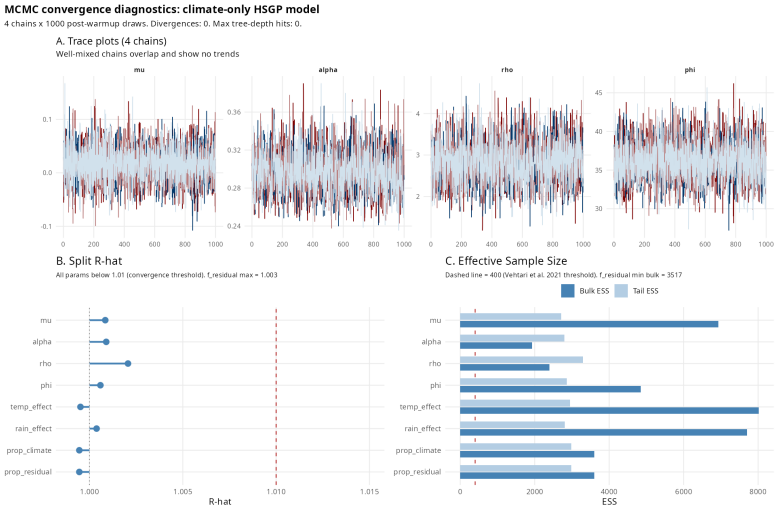
Decomposition is not an artefact of lag choice or GP hyperpriors.

Model Validation



- Posterior predictive tracks timing and magnitude of all major peaks (2013, 2019, 2020, 2022)
- **Observation layer is well calibrated**: empirical coverage of the nominal 95% interval $= 98.4\%$, 80% interval $= 91.8\%$
- This is not forecasting. This is a one-**week**-ahead predictive check: each week is predicted from the previous 6 weeks of **observed** cases (the generation interval support).

MCMC Convergence Diagnostics



Sampler: 4 chains $\times$ 1000 post-warmup draws (HMC/NUTS, `adapt_delta` $= 0.95$)

Diagnostics:

- **0** divergent transitions
- **0** max-tree-depth hits
- $\hat{R} < 1.01$ for every parameter and every f_{residual} component (max $= 1.003$)
- Bulk ESS > 1900 for all parameters
- Bulk ESS for $f_{\text{residual}} > 3647$
- Trace plots show well-mixed, overlapping chains with no drift

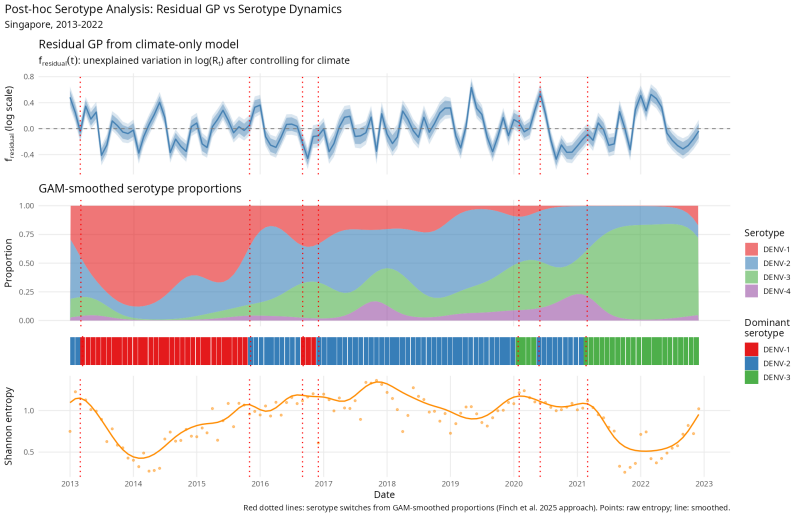
Post-hoc Serotype Analysis: Idea and Methods

Idea. The residual GP f_{residual} absorbs $\sim 99.4\%$ of R_t variance — unexplained by climate. We hypothesize that *serotype dynamics* (replacement, diversity) drive this residual via immunity depletion: a population encountering a newly dominant serotype is effectively more susceptible, elevating R_t.

What we do. Use the *already-fitted* posterior f_{residual} as a proxy for non-climate drivers and correlate it with independent serotype data. *No refit, no additional Stan covariates.* Four complementary lenses on the same residual:

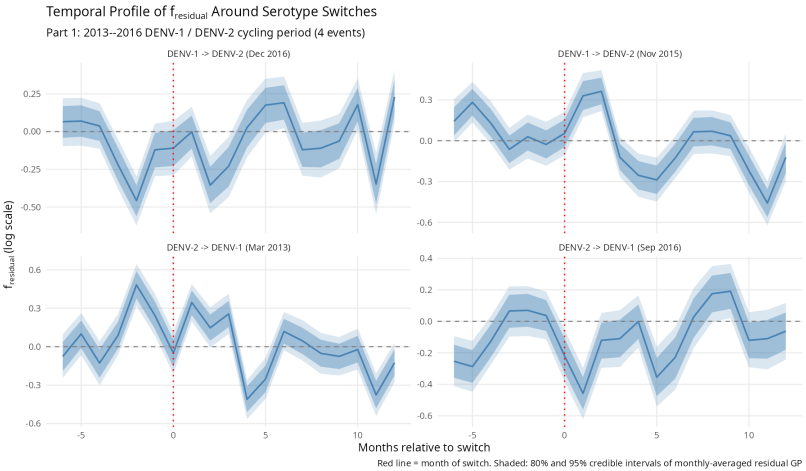
1. **Switch profiles** ($n = 7$ events)
   - Extract f_{residual} in $[-6, +12]$ month window around each switch
   - Tests event-by-event immunity-depletion signature (rise at switch, decay as immunity accumulates)
2. **Cross-correlation (CCF)**
   - Monthly f_{residual} vs. Shannon entropy and turnover rate, lags ± 12 months
   - Systematic lead–lag $\rightarrow$ early-warning window
3. **Spectral analysis**
   - Decompose f_{residual} into contributions at each cycle length (FFT power spectrum, credible band from draws)
   - Peaks flag the timescales on which the residual oscillates; tests whether they match the multi-annual cycle that cross-immunity + seasonal vector dynamics can generate (Wearing & Rohani 2006)
4. **Per-switch case decomposition**
   - Compare full model to climate-only reference $R_t^{\text{ref}} = \exp(\mu + f_{\text{climate}})$
   - Translates log-scale residual into per-event **case contribution** (retrospective accounting)

Serotype Switches Associate with Elevated R_t



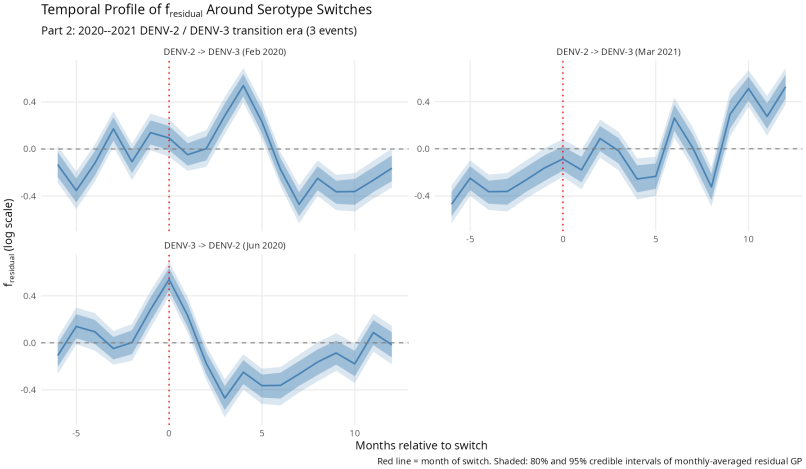
5 of 7 serotype switch events show $P(\text{elevated } f_{\text{residual}}) > 0.75$ in 0–3 months post-switch. Strongest: DENV-2 $\rightarrow$ DENV-3 (Feb 2020), $P = 1.0$, 20–35% R_t increase.

Temporal Profiles Around Switches: Part 1 (2013–2016)



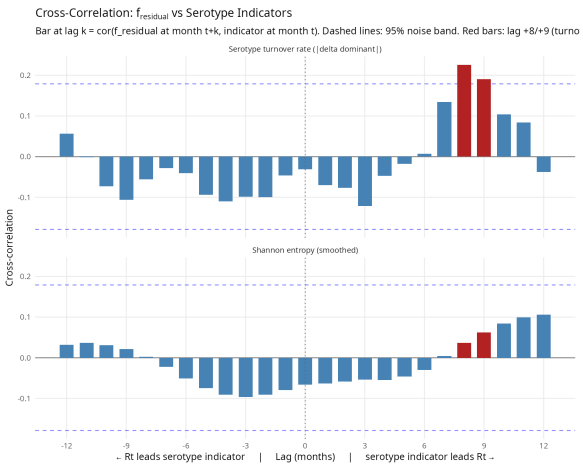
DENV-1 / DENV-2 cycling period. Predicted immunity-depletion pattern: **rise at switch, decline during stable dominance**.

Temporal Profiles Around Switches: Part 2 (2020–2021)



DENV-2 ↔ DENV-3 transition era. **Feb 2020 DENV-3 emergence** produces the clearest rise-and-decay signature (strongest event; $P = 1.0$).

Serotype Turnover Leads R_t by $\sim$8 Months



How to read the figure:

- Bar at lag $k = \mathrm{cor}(f_{\mathrm{residual},t+k},\ \mathrm{indicator}_t)$
- **Lag 0** = same month (contemporaneous)
- **Positive lag** = indicator leads R_t
- **Negative lag** = R_t leads indicator
- Dashed lines: $\pm 1.96/\sqrt{n}$ noise band

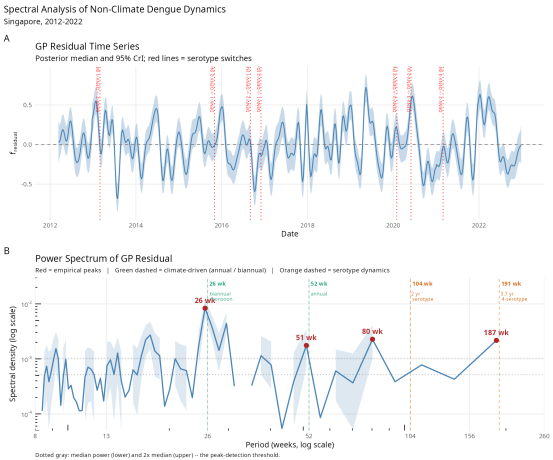
Indicators:

- *Turnover rate* = $|\Delta$dominant prop.$|$: month-to-month change in the dominant serotype's smoothed share
- *Shannon entropy*: diversity of co-circulating serotypes

Result:

- **Turnover rate** peaks at lag **+8 to +9 months** ($r \approx 0.23$ at $+8$, 0.19 at $+9$; lag $+8$ exceeds the noise band) $\Rightarrow$ turnover precedes R_t elevation by $\sim$8 months (red bars)
- *Entropy* CCF stays within the noise band at all lags

Spectral Analysis of f_{residual}: Structure Consistent with Serotype-Driven Immunity?



Reading the figure:

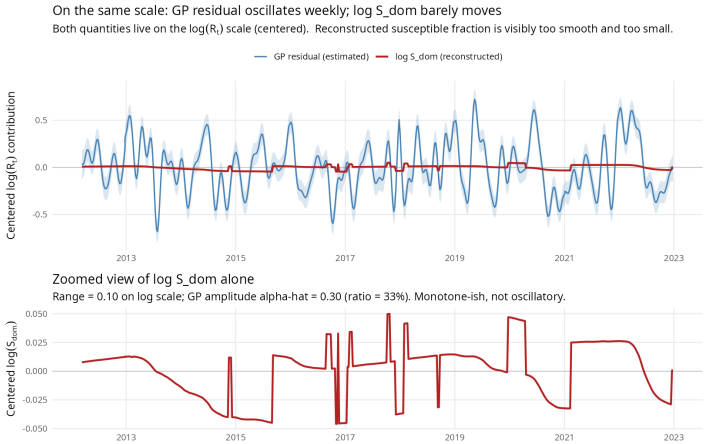
- **Top:** f_{residual} over time, switches in red
- **Bottom:** power spectrum: where the residual concentrates its variance, by cycle length

Peaks the spectrum reveals:

- **26 wks**: biannual monsoon
- **51 wks**: annual seasonality
- $\sim$1.5 **yr** (80 wk): multi-annual structure
- $\sim$3.6 **yr** (187 wk): multi-annual dengue cycle

Why this matters: the $\sim$3.6-yr peak sits in the range of the $\sim$3-year total-incidence cycle that cross-immunity plus seasonal vector dynamics can generate (Wearing & Rohani 2006). The residual is not white noise: it carries multi-annual structure consistent with cross-immunity dynamics.

Susceptible Depletion Cannot Generate the GP Residual



Serotype-aware susceptible reconstruction may not be what $f_{residual}$ is tracking.

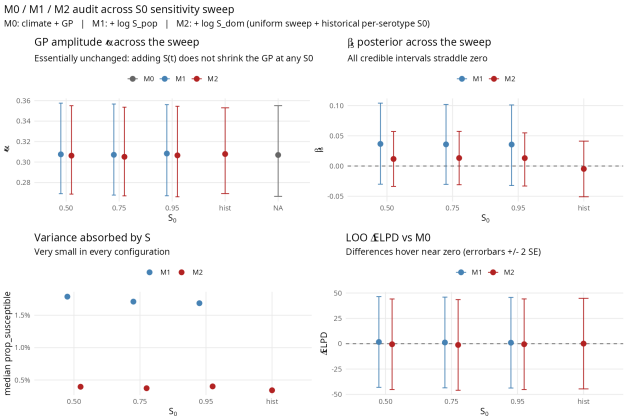
Question. If f_{residual} is really tracking serotype-driven immunity depletion, an explicit reconstruction of the susceptible fraction should look like it.

Method. Reconstruct $S_{\text{dom}}(t)$ = fraction susceptible to the *currently dominant* serotype, via per-serotype pools depleted weekly by their own cases, then compared on the same $\log R_t$ scale as f_{residual}.

Figure — two failures:

- **Top:** f_{residual} (blue) oscillates with amplitude $\hat{\alpha} \approx 0.30$; $\log S_{\text{dom}}$ (red) is essentially a flat line — total range only ≈ 0.10 ($\sim 33\%$ of the residual)
- **Bottom (zoom):** $\log S_{\text{dom}}$ on its own scale — jagged drops at dominance flips, but no high-frequency oscillation

Model Comparison: Adding Explicit Susceptibility Does Not Help



The residual is tracking something explicit susceptibility cannot capture: spatial heterogeneity, importations, vector dynamics, or other unobserved complexity.

Two susceptible covariates, each reconstructed weekly from cumulative cases under simple SIR-pool accounting:

- S_{pop} **(serotype-blind):** a *single* national pool depleted by *all* cases regardless of serotype — the classical SIR view
- S_{dom} **(serotype-aware):** *four* parallel pools, one per serotype, each depleted only by its own cases; at each week we pick the pool for whichever serotype is currently dominant

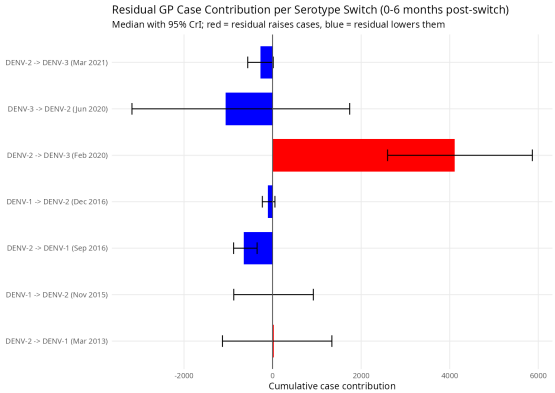
Three nested models (same priors, MCMC):

- **M0:** climate + GP (baseline)
- **M1:** M0 $+\beta_S \log S_{\text{pop}}$
- **M2:** M0 $+\beta_S \log S_{\text{dom}}$

Results (across full S_0 sensitivity sweep):

- GP amplitude $\hat{\alpha}$ **unchanged** across all 8 fits (0.305–0.308): adding S does *not* shrink the GP.
- β_S credible intervals **include zero** in every M1 and M2 configuration — including the data-driven historical M2.

Per-switch Residual Case Contribution



Question: of this week's expected cases, how many are attributable to f_{residual} rather than climate alone?

Setup — the renewal equation gives expected cases as

$$\lambda_t = R_t \cdot I_t, \qquad I_t = \sum_{s=1}^{6} c_{t-s}\, w_s$$

where I_t is the infectious pressure.

Method (3 steps):

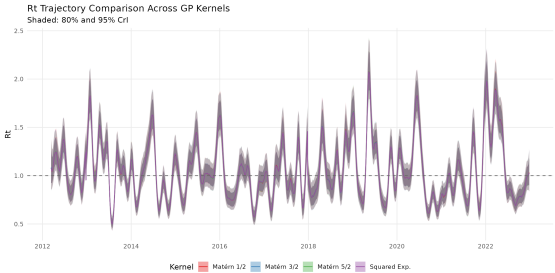
1. **Full model:** $\lambda_t = \exp(\mu + f_{\text{climate}} + f_{\text{residual}}) \cdot I_t$
2. **Climate-only reference:** $\lambda_t^{\text{ref}} = \exp(\mu + f_{\text{climate}}) \cdot I_t$ (zero out f_{residual}, keep *same* I_t)
3. **Cumulate the weekly gap** over the 6 months after each switch: $\sum_{t=0}^{26\text{ wk}} (\lambda_t - \lambda_t^{\text{ref}})$

Why I_t is the amplifier: a $+0.2$ on log R_t during a quiet week (small I_t) adds a handful of cases; the same bump during an outbreak (large I_t) adds thousands.

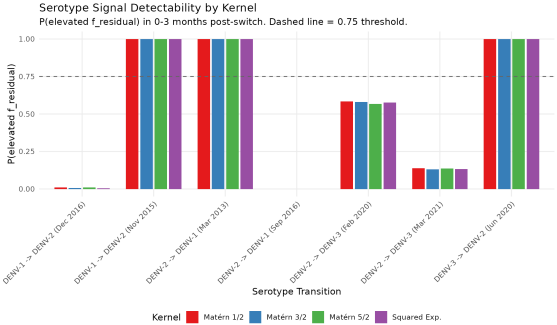
Feb 2020 DENV-3: +4,419 [2,925–6,043] = model expected ~4,400 more cases over 6 months post-switch than the climate-only reference. Other switches: ≈ 0.

Caveat: this is a *retrospective accounting* of how many of the observed cases the residual GP explains, not a *forward counterfactual* of how many cases the switch caused.

Kernel Comparison: Conclusions Are Robust

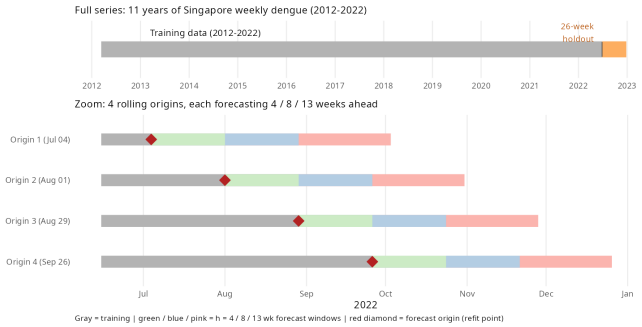


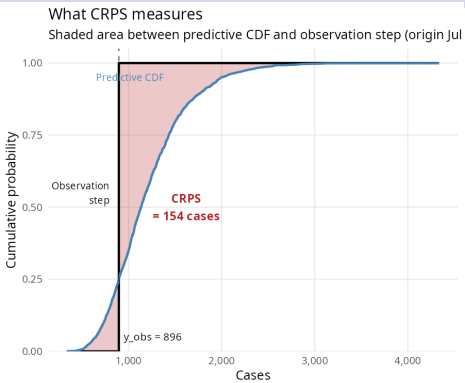
R_t trajectories nearly identical across all 4 kernels.
$|\Delta\text{ELPD}| < 2$ for all.



All 4 kernels detect the **same 5/7 switches** ($P > 0.75$).
Scientific conclusions hold regardless of kernel choice.

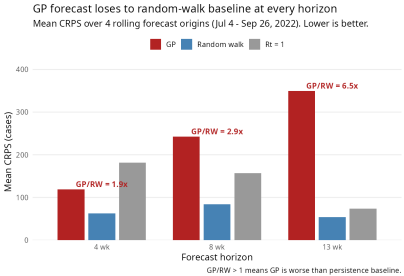
Forecasting: Methodology

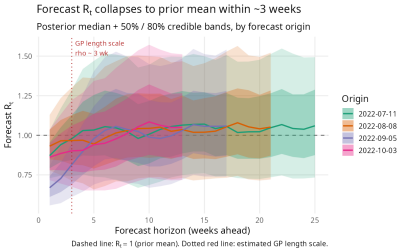




- **CRPS** (Continuous Ranked Probability Score) $= \int \left(F_{\text{pred}}(x) - \mathbf{1}\{x \geq y_{\text{obs}}\}\right)^2 dx$: integrated squared gap between the predictive CDF and the step function at the observed value (shaded area in right figure).
- **Baselines:** random walk on cases (persistence; min. bar) and $R_t = 1$ (endemic).
- **Skill** $= 1 - \text{CRPS}_{\text{GP}}/\text{CRPS}_{\text{baseline}}$ ($> 0 \Rightarrow$ GP wins).
- Caveat: 4 origins, late-2022 only. Illustrative experiment.

Forecasting: The GP Has No Predictive Skill





Top: GP loses to RW everywhere

- **1.9×, 2.9×, 6.5× worse** (4, 8, 13 wk)
- Skill vs RW: -7.6, -2.3, -10.5
- Worse than $R_t = 1$ beyond 4 wk

Bottom: R_t collapses to the mean

- Median $\to 1$ within $\sim$3 wk ($\approx \hat{\rho}$)
- Past $\hat{\rho}$, posterior = prior

Why: GP is a retrospective smoother, not a forward model.

Interpretation: operational forecasting needs mechanistic structure.

Branching Process: From R_t to Outbreak Risk

Question: R_t is an *average*. If one case arrives this week, what is the probability the chain *sustains* rather than going extinct by chance?

Offspring: each case produces $Y \sim \text{NB}(\text{mean}{=}R_t,\ \text{dispersion}{=}k)$ secondary infections (NB allows superspreading).

Extinction equation: let q = probability that a lineage from one case dies out. Galton–Watson: q is the smallest root in $[0,1]$ of $q = G(q)$, where $G(s) = E[s^Y]$ is the PGF:

$$G(s) = \left(1 + \tfrac{R_t}{k}(1-s)\right)^{-k}$$

$R_t \le 1 \Rightarrow q = 1$; $R_t > 1 \Rightarrow q \in (0,1)$ (solved numerically).

Outbreak risk: $P(\text{sustain}) = 1 - q$, computed per posterior draw of (R_t, ϕ) at every week, so R_t uncertainty flows through. We use $k \leftarrow \phi$ as a pragmatic proxy.

Caveat: *$P(\text{sustain})$ is a lens on potential risk, not a calibrated outbreak forecast.*

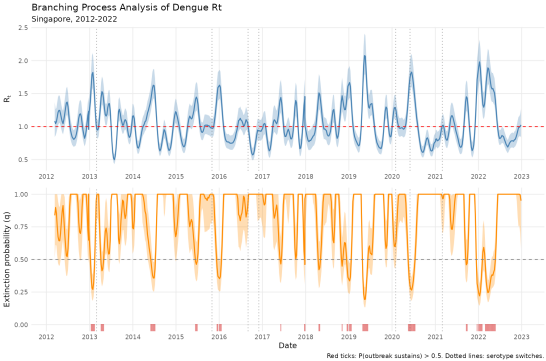
Why it matters:

- Non-linear near $R_t{=}1$: small shifts flip risk from 0
- Turns R_t into a vulnerability map
- Tests if switches open vulnerability windows

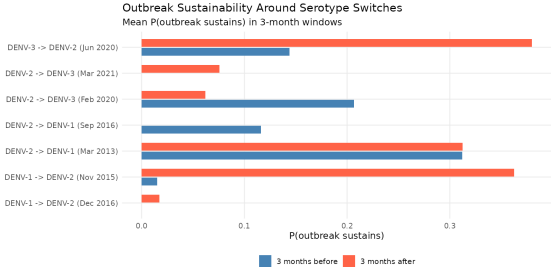
Assumptions of Galton-Watson that depart from dengue:

- Homogeneous mixing: ignores spatial clustering around breeding sites or households
- Independent offspring
- Infinite susceptible pool
- No antibody-dependent enhancement or cross-immunity between the 4 DENV serotypes

Branching Process: Vulnerability Windows



Strong anticorrelation: $\rho_{\text{Spearman}} = -0.94$ between R_t and extinction probability.



5/7 switches show increased $P(\text{sustain})$ post-switch (mean: $+6$ pp). 2022: 18 weeks of $P(\text{sustain}) > 0.5$.

Limitations

Methodological:

1. **GP dominance is partly structural.** With $\hat{\rho} \approx 3$ wks the GP can absorb *any* smooth variation, so the 99.4% residual share is *consistent with* serotype-driven dynamics but does not *prove* them. "Residual" means non-climate, not "serotype."
2. **Climate is scale-dependent.** $<1\%$ share at weekly resolution $\neq$ climate is unimportant overall — it likely acts through seasonal vector-ecology timescales a weekly GP is not built to resolve.
3. **GP has no forward mechanism.** The GP smooths *past* R_t; it contains no vector dynamics, immunity, or importations to propagate forward. Beyond $\sim\hat{\rho} \approx 3$ wks, the posterior collapses to the prior mean $R_t \approx 1$ and the forecast stops carrying information. *Structural, not a tuning issue.*
4. **Generation interval is fixed.** Built via Monte Carlo convolution at point estimates; its uncertainty is not propagated into R_t.

Epidemiological & data:

5. $n = 7$ **switch events.** Descriptive, pattern-based evidence (post-switch elevation, $\sim$8-mo CCF lead); no formal hypothesis testing possible.
6. **Constant case ascertainment assumed.** Cancels in the Cori renewal-equation framework, but Tan et al. 2019 report serotype-dependent infection-to-notification ratios of 3:1 to 16:1, which could bias post-hoc serotype interpretation.
7. **Spatial & serotype aggregation.** National, single-pool model. The susceptible-sensitivity sweep already showed an aggregate SIR pool *cannot* reproduce the residual amplitude — so local heterogeneity and serotype-specific immunity are doing real work.
8. **11-year window.** The $\sim$3.6-yr spectral peak spans fewer than 3 complete cycles in our data — the theoretical multi-serotype signature is suggestive, not established.

Conclusions & Future Directions

Key takeaways:

1. Climate explains $<1\%$ of weekly R_t variance; the residual GP captures $\sim 99.4\%$
2. Serotype switches associate with elevated R_t (5/7 events); serotype turnover rate leads R_t by ~ 8 months
3. **The residual tracks complex unobserved processes that explicit susceptibility cannot reproduce.** Maybe the residual is tracking spatial heterogeneity, importations, vector dynamics, or finer immune complexity beyond simple SIR-pool accounting
4. Decomposition is robust across climate lags, GP hyperprior means, and 4 kernels
5. GP provides **no forecasting skill**: tends to "revert to the mean"

Future directions:

1. **Spatially explicit, multi-serotype SIR fitted at NEA cluster level**: the susceptible-pool sensitivity sweep shows aggregate immunity *cannot* produce the observed residual; maybe we need both serotype-specific compartments and spatial heterogeneity
2. Benchmark against deep generative baselines (VAE, transformer, foundation models) on matched feature sets

Thank you

Questions?

Code and data available in the submitted repository.

Key References

Methodology

- Cori A. et al. (2013). A new framework and software to estimate time-varying reproduction numbers during epidemics. *Am. J. Epidemiol.* 178(9):1505–1512.
- Riutort-Mayol G. et al. (2023). Practical Hilbert space approximate Bayesian Gaussian processes for probabilistic programming. *Stat. Comput.* 33:17.
- Abbott S. et al. (2020). Estimating the time-varying reproduction number of SARS-CoV-2 using national and subnational case counts. *Wellcome Open Res.* 5:112 (EpiNow2).

Dengue biology & Singapore epidemiology

- Chan M., Johansson M.A. (2012). The incubation periods of dengue viruses. *PLoS ONE* 7(11):e50972.
- Bhatt S. et al. (2013). The global distribution and burden of dengue. *Nature* 496:504–507.
- Wearing H.J., Rohani P. (2006). Ecological and immunological determinants of dengue epidemics. *PNAS* 103(31):11802–11807.
- Reich N.G. et al. (2013). Interactions between serotypes of dengue highlight epidemiological impact of cross-immunity. *J. R. Soc. Interface* 10(86):20130414.
- Rajarethinam J. et al. (2018). Dengue in Singapore from 2004 to 2016. *Am. J. Trop. Med. Hyg.* 99(1):204–210.
- Tan L.K. et al. (2019). Force of infection and true infection rate of dengue in Singapore. *Am. J. Epidemiol.* 188(8):1529–1538.
- Finch E. et al. (2025). Climate variation and serotype competition drive dengue outbreak dynamics in Singapore. *Nat. Commun.* 16:11364.